A gateway between the console and the CT scanner that refuses what Saudi regulation forbids.

Samaam reads the patient's renal function, checks the scan settings against Saudi regulation, and answers 403 with the article and the issuing authority. The decision is deterministic code, not a language model.

The request

Iohexol · 150 mL
CTDIvol 18.0 mGy
DLP 900 mGy·cm

The usual protocol. It looks normal on the console.



eGFR, computed with CKD-EPI
19.1 threshold 30

Contrast-associated AKI risk in this band
≈30%

Decision in the recorded demo
29 ms observed, not a guarantee

The problem · SC-02, a simulated case

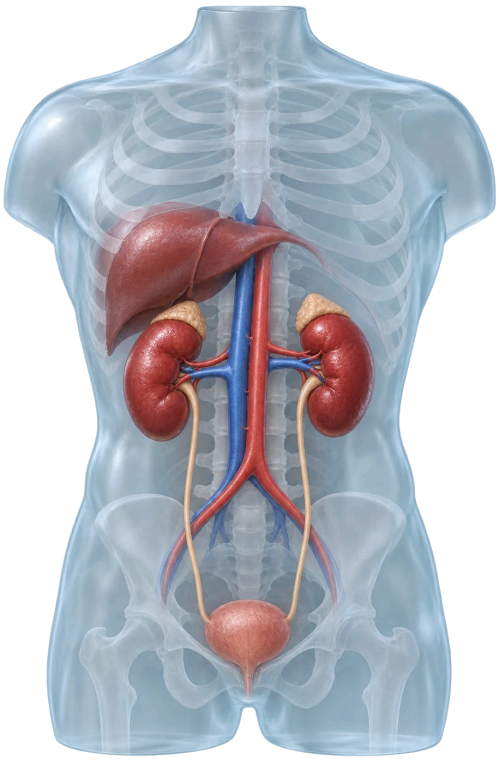
The protocol looks normal on the console

A woman of 62 with metastatic colorectal cancer is booked for a contrast CT of the chest, abdomen and pelvis, to stage her disease and check treatment response. Her serum creatinine is 229.8 µmol/L. She takes oxaliplatin, metformin 1000 mg twice daily, and ibuprofen when needed.

On a night shift the technologist enters the usual adult protocol: CTDIvol 18.0 mGy, DLP 900 mGy·cm, Iohexol 150 mL, no prophylaxis, metformin not held. Nothing between that keyboard and the scanner reads her renal function, and nothing checks the request against Saudi regulation.

Computed from her creatinine, her eGFR is 19.1. The Ministry of Health protocol puts the risk of contrast-associated kidney injury in that band at about 30%, indicates prophylaxis below 30, and requires metformin to be held.

This patient is synthetic. The numbers come from the project's scenario file, reviewed by a consultant radiologist.



What Samaam does

Four checks, each with its reading, its limit and its basis

Samaam computes eGFR itself, with the CKD-EPI equation the Ministry of Health protocol prints, and does not import the number. Direct identifiers are dropped before anything else sees the record. Each rule then cites its knowledge-base record by explicit ID, so the legal basis cannot drift with the wording of a query. For SC-02 the answer is `403`, verdict `VIOLATION`, action `AUTHORISE`, and the device stays locked.

Check	SC-02 reading	Basis and source
National reference level	`CTDIvol 18.0 > 14 mGy` `DLP 900 > 706 mGy·cm`	`STATUTORY` SFDA MDS-G008, binding by Royal Decree 60057. Raises `AUTHORISE`.
Renal prophylaxis	`eGFR 19.1 < 30` `none ordered`	`NATIONAL_PROTOCOL` Ministry of Health contrast protocol, section 5.2, p. 22. Raises `AUTHORISE`.
Metformin	`Category II, not held`	`NATIONAL_PROTOCOL` Section 5.3, p. 28: stop at the exam, suspend 48 hours. Raises `CONFIRM`.
Nephrotoxic medication	`oxaliplatin, ibuprofen`	`NATIONAL_PROTOCOL` A warning. Raises `CONFIRM`.

A binding law and a clinical protocol are not the same thing, and Samaam keeps them apart: a dose above the national level breaches a binding instrument, while missing prophylaxis departs from a Ministry of Health protocol and is never described as a legal breach. Samaam never blocks on contrast volume. Volume and rate follow the study, the patient's weight, the agent and the scanner protocol, so 150 mL is shown as an entered value only.

No model in the decision

If the model is deleted, every block still happens

The pipeline follows ITU-T Y.3172 node for node. The decision sits in P, plain rule logic. A language model only writes the explanation, about 40 seconds later in the recorded demo, from a separate endpoint and after the verdict has arrived.

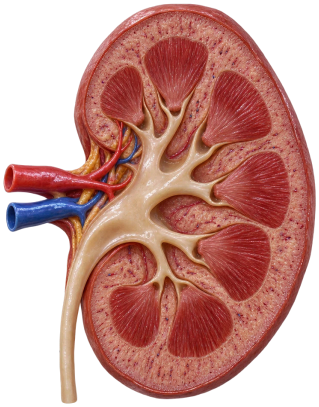
SRC	C	PP	M	P	D	SINK
Regulation, record, settings	DICOM MWL, HL7 FHIR	Drops identifiers, computes eGFR	Explains only	The decision, no model	Citations, audit	Device, interface

Why before exposure matters

A breach found afterwards has already happened

Saudi reference levels bind providers, and MDS-G008 requires automatic dose recording and periodic reports to the SFDA. That enforcement finds a breach after the patient was exposed. Commercial dose-management platforms are retrospective in the same way. NEMA XR 25 CT Dose Check does warn before the scan, but against preset values. It cannot see the patient.

> **CT Dose Check compares the machine to itself. Samaam compares it to the patient.**



Hold versus refuse

A clinical block is a hold. A privacy breach is a refusal.

AUTHORISE

SC-02 is held for a named doctor

No eGFR value forbids iodinated contrast outright. The Ministry of Health protocol makes stage 4 to 5 disease a relative contraindication and forbids withholding contrast from a life-threatening indication on renal grounds. Contrast may still be indicated in metastatic disease, and the radiologist decides.

A named consultant may release the scan. The release is recorded as `EXECUTED_UNDER_OVERRIDE`.



PROHIBITED

SC-03 is refused without exception

A service account asks to export 4,200 oncology records abroad for a marketing campaign. Three statutory PDPL checks fail: minimisation (Article 23), lawful basis, and cross-border transfer (Article 29). The session ends and the incident goes into the audit trail.

A clinician may accept clinical risk for their own patient. None may authorise what the PDPL forbids, so the same consultant name is refused here.



A consultant radiologist's review turned absolute blocks into holds for authorisation. The verdict is the regulatory classification. The action is what the console does:

PROCEED	CONFIRM	AUTHORISE	PROHIBITED
Nothing stands in the way	The operator acknowledges a risk factor	A named radiologist reviews and approves	No clinical override path exists

How it attaches

The scanner already asks. Samaam answers.

No CT scanner exposes an API to be driven through. It asks, over DICOM, for the work scheduled on it, and whoever answers decides what appears on the technologist's screen. Samaam is that answer. The device is not modified, updated or replaced.

Hospital system
patient, creatinine

HL7 FHIR R4 →

Samaam
evaluates every entry

← DICOM C-FIND

CT scanner
asks for its worklist

DICOM Modality Worklist

Samaam is the worklist server. A held study is not served, so it never appears on the console. Both outcomes go to the audit trail.

HL7 FHIR R4, read-only

Pulls the patient and the latest serum creatinine. Name, contact and address fields are never read, and nothing is written back.

The honest limit. This blocks the sanctioned path. It does not cut power, and a technologist keeps manual, unscheduled entry. That is deliberate: CT Dose Check is itself an alert requiring acknowledgement, because locking a scanner on a critical patient can kill them.

Proof

Three scenarios, reproducible in one command

Run `python run_demo.py` from the public repository. The three cases return these results against a mock device.

Case	Request	Verdict and action	Device
SC-01	Chest CT, eGFR 81.4, dose within national levels	COMPLIANT · PROCEED	**200** released
SC-02	Abdomen and pelvis, eGFR 19.1, DLP 900, metformin not held	VIOLATION · AUTHORISE	**403** locked, held
SC-03	Bulk export of 4,200 oncology records abroad for marketing	VIOLATION · PROHIBITED	**403** session ended
Empty	A request with no evidence	INSUFFICIENT_EVIDENCE	Samaam does not guess

45
regulatory records, 43 verified. Unverified records are never cited in a block.

1,338
source passages indexed from 7 primary documents

7
deterministic rules, 4 on the device path and 3 on the data path

6
regulatory gaps documented, each with a recommendation to the regulator

Every citation resolves to the document itself, not a ministry homepage. Sources include the Ministry of Health contrast protocol, SFDA MDS-G008, the PDPL and its implementing regulation, and ITU-T Y.3172. All patient data is synthetic. No real scanner or hospital integration exists yet. 29 ms is what the recorded demo showed, not a guarantee. The six regulatory verdicts are defined, and the current rules emit three of them. Medical renders are illustrations.



The award

3rd Prize, an international award from ITU

ITU AI Readiness Hackathon, Kingdom of Saudi Arabia, Health & Radiology track. Awarded to Team Samaam on 15 September 2026, signed by Seizo Onoe, Director of the ITU Telecommunication Standardization Bureau.

Team Samaam: Mohammed Nadher Abo Alrejal (Team Leader), Badour Alsehli, Haneen Saleh Alharbi, Norah Abdullah Alsahli.

The next step

From a mock device to one real worklist

- Test the gateway against a real worklist in one radiology department, under a consultant radiologist's review.
- Have clinicians review the thresholds and the wording of every hold on site.
- Prepare the SFDA path, where Rule 10 places software affecting radiology devices in Class C.



Code: github.com/aboalrejal-ai/samaam · Contact: aboalrejal.ai@gmail.com